WORKED SOLUTION

STEP III 2008, Question 13

Probability and Statistics • Sums of Indicator Variables

STEP III 2008, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q13

This question has a full guided treatment in *Beyond STEP*, Section 12.10, *Tying Loose Ends*: https://beyondstep.keithmcnulty.org/ch12/loose-ends.html

The key idea

Whatever happens at a step, the number of pieces of string always drops by exactly one: either a ring is removed from play, or two pieces become one. So after every step we are back in the same situation with one fewer piece, whatever happened before. That lets us write the total number of rings as a sum of *independent* indicator variables, and then the mean and variance are simply sums.

Solution

The first step

With n pieces there are $2n$ free ends, so there are

$$\binom{2n}{2} = n(2n-1)$$

equally likely ways to choose the two ends. A ring is formed exactly when both ends belong to the same piece, and there are n such pairs (one per piece). So

$$\mathrm{P}(\text{ring at the first step}) = \frac{n}{n(2n-1)} = \frac{1}{2n-1}.$$

Equivalently: whichever end we pick first, exactly one of the remaining $2n-1$ ends is its partner. If X_n is 1 when the first step makes a ring and 0 otherwise, the expected number of rings at the first step is

$$\mathrm{E}(X_n) = \frac{1}{2n-1}.$$

Expected number of rings

After the first step there are $n-1$ pieces of string, each with two free ends. This is true in both cases: a ring takes one piece out of play, and joining two different pieces turns two pieces into one. The lengths of the pieces don't matter; only their ends do.

> **Ask yourself** Does it matter that some pieces are now longer than others, or that a ring was or wasn't made earlier?

No. The only thing that affects which pair of ends is chosen is how many pieces (and hence ends) remain. So when there are k pieces left, the step makes a ring with probability $\frac{1}{2k-1}$, regardless of all earlier steps.

Label the steps by the number of pieces present just before them, $k = n, n-1, \ldots, 1$, and let X_k be the indicator that a ring is made at that step. Then $\mathrm{P}(X_k = 1) = \frac{1}{2k-1}$, and the process stops after the step with $k = 1$ (which always makes a ring, as $X_1 = 1$). The total number of rings is

$$R = X_1 + X_2 + \cdots + X_n,$$

so by linearity of expectation

$$\mathrm{E}(R) = \sum_{k=1}^{n} \frac{1}{2k-1} = 1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n-1}.$$

Variance

For the variance we need more than linearity: we need the X_k to be independent. They are. Given any outcomes of the earlier steps, the step with k pieces still makes a ring with probability $\frac{1}{2k-1}$. So the conditional distribution of X_k given all earlier X's is the same as its unconditional distribution, and so the X_k are independent.

Each X_k is Bernoulli with $p_k = \frac{1}{2k-1}$, so $\mathrm{Var}(X_k) = p_k(1-p_k)$, and for independent variables the variances add:

$$\begin{aligned}\mathrm{Var}(R) &= \sum_{k=1}^{n} \frac{1}{2k-1}\left(1 - \frac{1}{2k-1}\right) = \sum_{k=1}^{n} \frac{2k-2}{(2k-1)^2} \\ &= \sum_{k=1}^{n} \frac{1}{2k-1} \; - \; \sum_{k=1}^{n} \frac{1}{(2k-1)^2}.\end{aligned}$$

The case $n = 40\,000$

Write $H_m = 1 + \frac{1}{2} + \cdots + \frac{1}{m}$. The sum of reciprocals of odd numbers up to $2n-1$ is all reciprocals up to $2n$, minus the even ones:

$$\sum_{k=1}^{n} \frac{1}{2k-1} = H_{2n} - \left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2n}\right) = H_{2n} - \tfrac{1}{2}H_n.$$

Using $H_m \approx \ln m$ for large m,

$$\mathrm{E}(R) \approx \ln(2n) - \tfrac{1}{2}\ln n = \tfrac{1}{2}\left(2\ln 2n - \ln n\right) = \tfrac{1}{2}\ln(4n).$$

For $n = 40\,000$ we have $4n = 160\,000 = 20^4$, so

$$\mathrm{E}(R) \approx \tfrac{1}{2} \ln \left(20^4\right) = 2 \ln 20 \approx 6.$$

So from 40 000 pieces of string we expect only about six rings.

Checks

- $n = 1$: one piece, one ring for certain. The formulas give $\mathrm{E}(R) = 1$ and $\mathrm{Var}(R) = 0$, as they should.
- $n = 2$: the first step makes a ring with probability $\frac{1}{3}$, and the last step always does, so R is 2 with probability $\frac{1}{3}$ and 1 otherwise. That gives mean $\frac{4}{3}$ and variance $\frac{2}{9}$, matching $1 + \frac{1}{3}$ and $\frac{2}{9}$ from the formulas.
- Enumerating every sequence of choices exactly for $n \leq 5$, and simulating $n = 6$, agrees with both formulas. For $n = 40\,000$ the exact sum is $\mathrm{E}(R) \approx 6.28$. The extra 0.29 over our estimate comes from the crude approximation $H_m \approx \ln m$ (see below).

Common slips

- Taking the ring probability to be $\frac{1}{n}$ (choosing two *strings*) instead of $\frac{1}{2n-1}$ (choosing two *ends*).
- Adding the variances without first justifying that the X_k are independent. Linearity is enough for the mean, but not for the variance.
- Approximating $\sum \frac{1}{2k-1}$ by $\ln n$ or $\frac{1}{2} \ln n$ without splitting off the even terms.

Beyond STEP

A sharper estimate is $H_m = \ln m + \gamma + o(1)$, where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. This gives $\mathrm{E}(R) \approx \frac{1}{2} \ln(4n) + \frac{1}{2}\gamma$, and also $\mathrm{Var}(R) \approx \frac{1}{2} \ln(4n) + \frac{1}{2}\gamma - \frac{\pi^2}{8}$, since $\sum 1/(2k-1)^2 = \pi^2/8$. The same "sum of independent indicators with probabilities $1/k$" structure counts the cycles of a random permutation and the records in a random sequence; in both cases the count grows like $\ln n$. Grimmett and Welsh, *Probability: An Introduction*, is a good place to go further with indicator variables and these random structures.

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